

SYLLABUS :-

Credits: 2 L-T-P: 0-0-3 Prerequisite: None Course Overview: This course is the second step in introducing constructional details to students. Based on the lectures delivered and reference drawings provided, the students shall prepare construction drawings and details. These lectures shall focus on constructional practices used in commercial and office buildings like curtain walls, partition walls, special flooring and false ceiling. The students will be encouraged to draw constructional details popularly used by vendors. The understanding of this course will act as a prerequisite for the next semester Working Drawing I and Working Drawing II in the subsequent semester. **Learning Objectives:** 1. Understand and illustrate fixing details of Aluminium sliding doors and windows. 2. Get acquainted with the fixing details and component parts of Aluminium partition walls, Aluminium glazed Curtain walls and Spider Joint Curtain walls. 3. Understand the constructional details of cavity walls and expansion joints at foundation, plinth, intermediate floor, terrace and parapet levels. 4. Examine the different types of terracing and water proofing treatments done on flat roofs and details of rain water pipe connections along with sectional details of basement walls, garden paving and site drainage. 5. Get acquainted with special types of vibration resistive flooring systems and its fixing details. 6. Recognize various fixing systems used for false ceilings while understanding fixing details of lighting fixtures and diffusers. **Course Curriculum:** Module 1 : Metal doors and windows Fixing details of Aluminium doors and windows, focus on sliding doors and windows. Module 2 : Partition and Curtain walls Fixing details and components of Aluminium partition walls, fully glazed, partly glazed or opaque panels. Fixing details and components of External Curtain wall, both Aluminium and Spider Joint types. Module 3 : Cavity wall and Expansion joints Typical section through a cavity wall showing details at roof, foundation and openings with appropriate waterproofing. Construction details of expansion joints at foundation, walls, floors and roof level for both concrete and brick work. - 35 - 30 | Department of Architecture and Regional Planning, IIT Kharagpur Module 4 : Terracing, basement wall, site drainage Water proofing and terracing treatment on flat roof with typical details of rain water pipe connections. Sectional details of basement walls. Garden paving and site drainage Module 5 : Special flooring Different types of special flooring, including resilient and vibration resistive floors like sports floors and flooring of industrial spaces. Module 6 : False Ceiling Suspended ceiling with different materials and finishing including fixing details of lighting fixtures and diffusers for ventilation and air conditioning. **Reading List:** 1. Allen, E. and Rand, P. (2016). Architectural Detailing. New Jersey: Wiley. 2. Arora, S.P. and Bindra, S.P. (2010). A Textbook of Building Construction, New Delhi: Dhanpat Rai Publications. 3. Bureau of Indian Standard (BIS), (2016). National Building Code of India 2016 (Volume-1 and 2). 4. Ching, F. D. K. (1991). Building Construction

Illustrated. New York, NY: Van Nostrand Reinhold. 5. Chudley, R. and Greeno, R. (2006). Advance Construction Technology. New Jersey: Prentice Hall. 6. Chudley, R. and Greeno, R. (2008). Building Construction Handbook. Oxford: Elsevier Ytytytyt 7. Emmitt, S. Olie, J. and Schmid, P. (2004). Principles of Architectural Detailing. New Jersey: Wiley. 8. Emmitt, S. and Gorse, C. (2010). Barry's Introduction to Construction of Buildings. New Jersey: Wiley Blackwell. 9. Emmitt, S. (2018). Barry's Advance Construction of Buildings. New Jersey: Wiley-Blackwell. 10. Frederick, S.M. and Jonathan, T. R. (2000). Building Design and Construction Handbook, McGraw-Hill 11. Gatz, K. (2000). Modern Architectural Detailing. New York: Van Nostrand Reinhold. 12. Mckay, W.B. (2004). Building Construction, (Volume 1 to 5), Routledge 13. Prabhu, B.T.S., Paul, K.V. and Vijayan, K. (2016), Calicut: Spades Publisher and Distributors.